Rohan Sinha

Last updated in August, 2026

Contact Information

720 Serra Street
Apt. 701
Stanford, CA
94305, USA
Tel: (510)-708-9690

Homepage: https://rohansinha.nl/
✉ E-mail: rhnsinha@stanford.edu
Linkedin: connect with me here
Twitter: click to follow me
Github: https://github.com/GoldeneyeRohan

Research Interests

My PhD research centers on increasing generalization and general-purpose robotic capabilities using foundation models, with a particular focus on 1) improving reliability and advancing AI safety for embodied systems, and on 2) data-centric methods to understand and improve robotics models through their training data quality.

Education

Stanford University 2020–2026
PhD. Candidate, Autonomous Systems Lab
M.S., Aeronautics and Astronautics
Advisor: Prof. Marco Pavone, (Rotation: Prof. Sanjay Lall)

University of California, Berkeley 2015–2020
B.A. Computer Science
B.S. Mechanical Engineering
Advisors: Prof. Benjamin Recht, Prof. Francesco Borrelli

Experience

Stanford University– Graduate Research Assistant 2020-2026
Advisor: Prof. Marco Pavone, Autonomous Systems Lab (ASL)
Thesis: "Towards Trustworthy Autonomy: Guardrails to detect, avoid, and learn from failures"
I work broadly on robot learning for manipulation and beyond, with an emphasis on reliability and safety under out-of-distribution (OOD) conditions. This involves developing both inference-time systems that catch and correct failures by leveraging frontier reasoning models and training-time algorithms that make robotic FMs, e.g., VLAs, more robust, along three trusts:

1. Runtime monitors and guardrails for embodied AI systems: *E.g., failure and misuse detectors, runtime monitors, and reasoning-based safety guardrails.*
2. Robot learning algorithms and systems that strengthen reliability and trustworthiness: *E.g., hierarchical policy architectures leveraging frontier models and inference-time scaling for VLAs.*
3. Towards a science of robot data: *E.g.,* Data selection and curation *algorithms to interpret and improve policies through their training data quality, towards durable deployment data-flywheels.*

My work was recognized with the **outstanding paper award** at RSS 2024 for a "thinking, fast & slow" reasoning architecture that enabled the real-time use of LLMs/VLMs to address the long tail of edge cases. I also received the **best paper award** at the 2025 RSS RoboEval workshop for an influence-function-based data curation method that substantially improved the performance and robustness of imitation-learned VLAs. Previously, I worked on deep Bayesian meta-learning and robust control.

Google DeepMind– Student Researcher 2024
Developed inference-time policy steering algorithms to steer vision-language-action (VLA) policies towards preferred behavioral modes and away from failures. Enhanced the out-of-distribution deployment reliability and human preference alignment of the state-of-the-art Gemini Robotics VLA in the context of the DeepMind robotics AI safety efforts. Work completed under Dr. Sumeet Singh and Dr. Vikas Sindhwani.

University of California, Berkeley– Undergraduate Research Assistant 2018–2020
Advisor: Prof. Benjamin Recht, Berkeley AI Research Lab (BAIR) (2019–2020)
Research on learning-based control strategies from monocular visual sensing. Developed a model car that races autonomously at high speeds in cluttered environments from a single human demonstration.
Advisor: Prof. Francesco Borrelli, Model Predictive Control (MPC) Lab (2018–2019)
Research focused on data-driven control algorithms. Worked on Learning Model Predictive Control (LMPC) strategies with convergence to model-free learned strategies subject to stochastic disturbances, and LMPC for multi-agent autonomous racing.

Amazon Inc.– Software Development Engineering Intern 2018
Data engineering for Amazon Music's distributed search pipeline.

Motional Inc. (Formerly Delphi Automotive)– Autonomous Driving Engineering Intern 2017
Developed and validated high-fidelity vehicle dynamics models for the Delphi–BMW self-driving simulator, plus algorithms for system identification and state estimation.

Honors and Awards

Best Paper Award at the RSS Robot Evaluation for the Real World Workshop	2025
Outstanding (Best) Paper Award at the Robotics: Science and Systems Conference	2024
Honorable Mention, NSF Graduate Research Fellowship Program	2021
Best project award, AA228: Decision Making Under Uncertainty	2020
Honors at graduation, UC Berkeley Faculty of Mechanical Engineering	2020
Distinction in General Scholarship, UC Berkeley Faculty of Computer Science	2020
Best Project Award, ME102B: Capstone Mechatronics Design	2019
Third place, NASA Aeronautics Advanced Aerial Vehicles Student Challenge	2017
Tau Beta Pi Honor Society, UC Berkeley	2017
Pi Tau Sigma Honor Society, UC Berkeley	2017
Dean's List, UC Berkeley College of Engineering	2015–2019
Top 15 Finalist, National Physics Olympiad, the Netherlands	2015

Teaching

AA289/CS529: Robotics and Autonomous Systems Seminar – Organizer 2022-2026
Revived Stanford's weekly robotics seminar (150+ students, $30k annual budget), hosting academic and industry speakers on frontiers in robotics research with recorded talks accumulating over 300k+ views on YouTube. See course webpage.

AA174/CS137A: Principles of Robot Autonomy–Head Course Assistant 2024
Head TA for Stanford's Ugrad/MS intro robotics course covering trajectory optimization, motion planning, perception and estimation (e.g., SLAM), and ML (e.g., CNNs). I developed assignments, managed course logistics, and ran hardware sessions on turtlebots.

EE263: Introduction to Linear Dynamical Systems–Course Assistant 2020
Lead TA for graduate course in linear algebra and dynamical systems under Prof. Sanjay Lall. I was responsible for general course logistics, held weekly discussion sections and office hours, wrote homework assignments and exams, and managed the team of graders for the course.

CS61B: Data Structures–Lab Assistant 2018
Lab Teaching Assistant for Computer Science 61B at UC Berkeley. CS61B is a fundamental software engineering course covering OOP, data structures, introductory graph theory, and other essential programming concepts.

Service and Leadership

Workshop Organizer – 1st Workshop Making Sense of Data in Robotics: Composition, Curation, and Interpretability at Scale, held at the 2026 Conference on Robot Learning (CoRL).

Principal Workshop Organizer – 2nd Workshop on Out-of-Distribution Generalization in Robotics: Towards Trustworthy Learning-based Autonomy, held at the 2025 Robotics: Science and Systems conference.

Instructor – "Test of AI and Emerging Technologies" Course – I helped develop and instruct a new course on AI and autonomy for the United States Air Force Test Pilot School. (2025)

Workshop Organizer – Workshop on Safely Leveraging Vision-Language Models in Robotics: Challenges and Opportunities, held at the 2025 International Conference on Robotics and Automation.

Graduate Steering Committee Member, Stanford Robotics Center – Stanford University (2023-present)

Principal Workshop Organizer – Workshop on Out-of-Distribution Generalization in Robotics: Towards Reliable Learning-based Autonomy, held at the 2023 Conference on Robot Learning.

President, Dutch Student Association @ Stanford – Stanford University, previously vice-president from 2022-2023 (2023-present)

Mentor, Stanford Robotics Center Summer Research Program – Mentoring first generation, low income, and racially minoritized students with the goal of providing real-world experiences in science and engineering research (2023)

Organizer, Dutch Royal Visit – I was key in organizing a state visit by H. M. Queen Maxima of the Netherlands to Stanford, accompanied by a trade delegation comprising Dutch cabinet ministers, industry and nonprofit executives, and academic leaders: I organized a seminar on autonomous driving, a panel on AI, a student lunch with the education minister, and an autonomous vehicle showcase that was extensively covered in the media.

Reviewer – IEEE TCST (2021), WAFR (2022), IEEE CCTA (2022), IEEE CDC (2023, 2024), AAAI (2023, 2024), RSS (2025), IROS (2025), CoRL (2025), RA-L (2025)

Diversity, Equity, and Inclusion Committee Member – Stanford Aero/Astro (2020–2023)

President, Goldeneye Student Engineering Team – UC Berkeley (2017–2019)

Officer, California Sailing Team – UC Berkeley (2016–2017)

Invited Talks

Stanford Robotics Seminar (May 2025)
TU Munich Cyber Physical Systems Group (December 2025)
Keynote, RSS Workshop on Statistical Uncertainty Quantification in the Era of AI-Enabled Robots (June 2025)
Annual meeting of the NASA University Leadership Initiative, Safe Aviation Autonomy (June 2025)
Keynote, Coordinated Science Conference, UIUC ECE (February 2025)
Toyota Research Institute (TRI) (October 2024)
DARPA TIAMAT Program Meeting (October 2024)
KTH Royal Institute of Technology Stockholm (October 2024)
Interactive Perception and Robot Learning Lab (June 2024)
Google DeepMind Robotics (June 2024)
Annual meeting of the NASA University Leadership Initiative, Safe Aviation Autonomy (June 2024)
Blue Origin (January 2024)
Center for Automotive Research at Stanford's annual affiliate meeting (November 2023)
Toyota Research Institute (September 2023)
Apple Inc., Special Projects Group ML seminar (September 2023)
Keynote at IEEE Space Mission Challenges for Information Technology - IEEE Space Computing Conference, workshop on Addressing Trust Challenges in Space Autonomy (July 2023)
Stanford Center for Automotive Research industry affiliate seminar (July 2023)
Annual meeting of the NASA University Leadership Initiative, Safe Aviation Autonomy (June 2023)

Publications

Journal Publications:

1. **Sinha, R.**, Dixit, A., Tian, R., Bajcsy, A., Majumdar, A., (2026). "Rethinking Safety for Generalist Robots". In: *Science Robotics*. (submitted) arXiv preprint arXiv:2609.06326)

2. Elhafsi, A., **Sinha, R.**, Agia, C., Schmerling, E., Nesnas, I. A. D., Pavone, M., (2023a). "Semantic Anomaly Detection with Large Language Models". In: *Autonomous Robots*. Special Issue on Large Language Models in Robotics

3. **Sinha, R.**, Sharma, S., Banerjee, S., Lew, T., Luo, R., Richards, S. M., Sun, Y., Schmerling, E., Pavone, M., (2022). "A System-Level View on Out-of-Distribution Data in Robotics". In: *arXiv preprint arXiv:2212.14020*

4. **Sinha, R.**, Harrison, J., Richards, S. M., Pavone, M., (2022a). "Adaptive Robust Model Predictive Control via Uncertainty Cancellation". In: *IEEE Transactions on Automatic Control*. (Accepted). Available at https://arxiv.org/abs/2212.01371

Conference Publications:

1. **Sinha, R.**, Singh, S., Shah, D., Harrison, J., Majumdar, A., Sindhwani, V., (2026). "Inference-Time Steering of Diffusion Models with Control Barrier Functions". In: (in preparation)

2. Bauer, E., Thumm, J., Agia, C., **Sinha, R.**, Xu, C., Nishimura, H., Bohg, J., Pavone, M., Itkina, M., (2026). "Behavior Graphs: Unsupervised Graph Structures for Interpreting Robot Deployment Data". In: *Proceedings of the 2027 IEEE International Conference on Robotics and Automation (ICRA)*. (submitted)

3. Marchiori, F., **Sinha, R.**, Agia, C., Robey, A., Pappas, G. J., Conti, M., Pavone, M., (2026). "Preventing Robotic Jailbreaking via Multimodal Domain Adaptation". In: *Proceedings of the 2026 IEEE International Conference on Robotics and Automation (ICRA)*

4. Christensen, K. A., Tufte, A. G., Gusev, A., **Sinha, R.**, Ganai, M., Alsos, O. A., Pavone, M., Steinert, M., (2026). "Foundation models on the bridge: Semantic hazard detection and safety maneuvers for maritime autonomy with vision-language models". In: *Ocean Engineering*

5. Agia, C., **Sinha, R.**, Yang, J., Antonova, R., Pavone, M., Nishimura, H., Itkina, M., Bohg, J., (2025). "CUPID: Curating Data your Robot Loves with Influence Functions". In: *9th Annual Conference on Robot Learning*. **(Best Paper Award @ RSS RoboEval Workshop, 2025)**

6. Kwok, J., **Sinha, R.**, Agia, C., Foutter, M., Li, S., Stoica, I., Mirhoseini, A., Pavone, M., (2025). "RoboMonkey: Scaling Test-Time Sampling and Verification for Vision-Language-Action Models". In: *9th Annual Conference on Robot Learning*

7. Ganai, M., **Sinha, R.**, Agia, C., Morton, D., Pavone, M., (2025). "Real-Time Out-of-Distribution Failure Prevention via Multi-Modal Reasoning". In: *9th Annual Conference on Robot Learning*. **(Oral Presentation)**

8. Soroka, E., **Sinha, R.**, Lall, S., (Apr. 2025). "Learning Temporal Logic Predicates from Data with Statistical Guarantees". In: *Proceedings of the 7th Annual Learning for Dynamics & Control Conference*. Proceedings of Machine Learning Research. PMLR

9. Agia, C., **Sinha, R.**, Yang, J., Cao, Z., Antonova, R., Pavone, M., Bohg, J., (2024). "Unpacking Failure Modes of Generative Policies: Runtime Monitoring of Consistency and Progress". In: *8th Annual Conference on Robot Learning*

10. **Sinha, R.**, Elhafsi, A., Agia, C., Foutter, M., Schmerling, E., Pavone, M., (2024). "Real-Time Anomaly Detection and Reactive Planning with Large Language Models". In: *Robotics: Science and Systems*. **(Winner: Outstanding Paper Award (top 0.2%))**. URL: https://tinyurl.com/aesop-llm

11. **Sinha, R.**, Schmerling, E., Pavone, M., (2023). "Closing the Loop on Runtime Monitors with Fallback-Safe MPC". in: *Proc. IEEE Conf. on Decision and Control*. URL: https://tinyurl.com/fallback-safe-mpc

12. Luo, R., **Sinha, R.**, Hindy, A., Zhao, S., Savarese, S., Schmerling, E., Pavone, M., (2023). "Online Distribution Shift Detection via Recency Prediction". In: *Proc. IEEE Conf. on Robotics and Automation*. (Accepted)

13. **Sinha, R.**, Harrison, J., Richards, S. M., Pavone, M., (2022b). "Adaptive Robust Model Predictive Control with Matched and Unmatched Uncertainty". In: *American Control Conference*

14. Anand, R.*, English, A.*, Gao, D.*, Malekshahi, S.*, **Sinha, R.***, Stevenson, N.*, (2017). "Goldeneye AB1". In: *Presented at NASA Aeronautics Design Challenge*. **(Third place/honorable mention)**. URL: http://goldeneyerohan.github.io/files/GoldeneyeAB1.pdf. **equal contribution*

Workshop Papers:

1. Foutter, M., Bhoj, P., **Sinha, R.**, Elhafsi, A., Banerjee, S., Agia, C., Kruger, J., Guffanti, T., Gammelli, D., D'Amico, S., Pavone, M., (2024). "Adapting a Foundation Model for Space-based Tasks". In: *RSS'24 Workshop on Semantics for Robotics: From Environment Understanding and Reasoning to Safe Interaction*. URL: https://arxiv.org/abs/2408.05924

2. **Sinha, R.**, Dyro, R., Pavone, M., (2023). "Beyond Basics: X-Plane Aircraft Simulator Offers Real-World Testing Environment for Control Systems". In: *Workshop on Benchmarking, Reproducibility, and Open-Source Code in Controls at CDC 2023*. URL: https://github.com/StanfordASL/ASL-X-Plane-11-Platform

3. Foutter, M., **Sinha, R.**, Banerjee, S., Pavone, M., (2023). "Self-Supervised Model Generalization using Out-of-Distribution Detection". In: *First Workshop on Out-of-Distribution Generalization in Robotics at CoRL 2023*. URL: https://openreview.net/forum?id=z5XS3BY13J

4. Elhafsi, A., **Sinha, R.**, Agia, C., Schmerling, E., Nesnas, I. A. D., Pavone, M., (2023b). "Semantic Anomaly Detection with Large Language Models". In: *Robotics, Systems and Science; Workshop Towards Safe Autonomy: New Challenges and Trends in Robot Perception*

5. **Sinha, R.**, Lall, S., (2022). "Cautious Markov Games for Interaction Aware Robotics". In: *Conference on Robot Learning: Workshop on Strategic Multi-Agent Interactions*

Other Writing:

1. Brown, R.*, Dyro, R.*, **Sinha, R.***, (2021). *Covariate Shifts in Multi-Agent Interactions*. Tech. rep. Stanford CS329D. **equal contribution*

2. **Sinha, R.** (2021b). *Solving Multi-Agent Zero-Sum Games with Mirror Descent*. Tech. rep. Stanford EE364B

3. **Sinha, R.***, Soroka, E.*, (2021). *Pose Graph Optimization Using Matrix Sketching*. Tech. rep. Stanford AA273. **equal contribution*

4. **Sinha, R.** (2021a). *Multi-Vehicle Autonomous Racing with Learning MPC and Trajectory Forecasting*. Tech. rep. Stanford AA277

5. **Sinha, R.** (2020). *Cautious Markov Games, a Novel Framework for Human-Robot Interaction*. Tech. rep. **(Best Project Paper)**. Stanford AA228

6. Narang, A.*, **Sinha, R.***, Siththaranjan, A.*, Yang, F.*, (2019). *Data-Poisoning for Linear Models*. Tech. rep. UC Berkeley EE227B. URL: https://rohansinha.nl/portfolio/projects-4/. **equal contribution*

7. Anderson, J.*, Chang, B.*, Cosner, R.*, Kruger, S.*, Lim, R.*, **Sinha, R.***, (2019). *Oversized Load Lifting and Yielding (Project OLLY)*. tech. rep. **(Best Project)**. UC Berkeley ME102B. URL: https://rohansinha.nl/portfolio/projects-3/. **equal contribution*

8. Anand, R.*, Anderson, J.*, Lim, R.*, **Sinha, R.***, (2019). *MPC Control of Multiple Quadcopters Cooperatively Lifting an Object*. Tech. rep. UC Berkeley ME231A. URL: https://rohansinha.nl/portfolio/projects-2/. **equal contribution*

Ongoing Works In Progress:

1. Title TBD: VLM-based failure analysis and synthetic data generation for self-improvement
2. Title TBD: Scalable data curation, collection, and interpretability for VLA policies
3. Title TBD: Improving trustworthiness of reasoning-based robotics foundation models
4. Title TBD: Safe humanoid control policies

Mentoring

Erik Bauer (2026), currently CS PhD Candidate at Stanford
Jerry Wang (2026), currently CS undergraduate at Stanford
Anton Kuznietsov (2026), currently PhD candidate at TU Darmstadt
Kim Christensen (2025), currently PhD candidate at NTNU
Francesco Marchiori (2025), currently PhD candidate at Univ. of. Padova
Milan Ganai (2025-current), currently CS PhD candidate at Stanford
Cassie Chen (2025), currently ME MS candidate at Stanford
Kai Storms (2024-2025), currently PhD candidate at TU Darmstadt
Jacky Kwock (2024-current), currently CS PhD candidate at Stanford
Sidharth Tadeparti (2024-2025), ME MS at Stanford, currently ML Eng. at Dexterity AI
Andrew Wang (2023-2025), AA MS at Stanford, currently controls at Tesla Optimus
Alex Wang (2024), currently CS BS student at Stanford
Xinmeng Zheng (2024), currently ICME PhD candidate at Stanford
Anish Muthali (2023-2024), currently quantitative researcher at Headlands technologies
David Shen (2023-2024), currently ML Eng. at LinkedIn
Matthew Foutter (2023), currently ME PhD candidate at Stanford
Gabriel Escatel (2023), currently BS candidate at Foothill Community College
Shahir Rahman (2022), currently CS Ph.D. candidate at Stanford

Skills and Interests

Languages: English (fluent), Dutch (native), French (proficient), German (elementary).
Technical Skills: `Python`, `Java`, `C/C++`, `Matlab`, `ROS`.
Sports/Hobbies: Sailing, Tennis, Soccer, Field Hockey, Snowboarding.